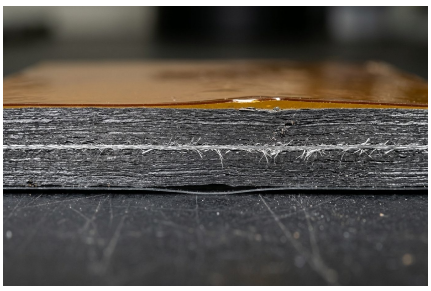
GRAPHITE-FLEX SERIES (GFlex-1000 / GFlex-1500)

InsulTherm Graphite-Flex™

5-Layer Edge-Sealed Pyrolytic Graphite Lateral Heat Spreader (ASTM E595 Low-Outgassing)

InsulTherm Graphite-Flex™ is an ultra-thin, 5-layer carrier-backed synthetic pyrolytic graphite laminate engineered to eliminate intense localized processor, GaN/RF, and optical hotspots by spreading heat laterally (1,000–1,500 W/m·K in-plane—up to 4× pure copper at <25% weight). Integrating a neutral-axis woven E-Glass scrim, an edge-overlapping 0.025 mm Kapton® dielectric cap (>5 kV), and an ASTM E595 silicone-free adhesive, it prevents carbon flaking and optical lens fogging.



Macro Cross-Section / Studio Hardware Inspection

1,000–1,500 W/m·K	ASTM E595 (TML <1%)	>5 kV Kapton Cap	Silicone-Free	GTE Ban Compliant

1. CHEMICAL / MATERIAL COMPOSITION & INTERNAL LAYER ARCHITECTURE

Polymeric precursor film carbonized above 2,500°C into oriented hexagonal SP2 carbon planes. Dual pyrolytic graphite cores are laminated symmetrically across an ultra-thin woven E-Glass fiberglass neutral-axis scrim using a 100% siloxane-free ASTM E595 acrylic PSA (TML < 1.0%, CVCM < 0.1%) beneath a 0.025 mm translucent amber Kapton® polyimide dielectric shield.

Layer 1 (0.025 mm)	Translucent Amber Kapton® Polyimide Dielectric Top Cap (Edge-Sealed, >5 kV)
Layer 2 (Upper Core)	High-Crystallinity Oriented Pyrolytic Graphite Matrix (k_xy = 1,000–1,500 W/m·K)
Layer 3 (Center Axis)	Embedded Woven E-Glass Fiberglass Scrim + ASTM E595 Bonding Line
Layer 4 (Lower Core)	Balanced Lower Synthetic Pyrolytic Graphite Spreader Matrix
Layer 5 (0.050 mm)	Clear Siliconized PET Release Liner + 0.010 mm Silicone-Free Acrylic PSA

2. TYPICAL PHYSICAL, THERMAL & ELECTRICAL PROPERTIES

PROPERTY / PARAMETER	TYPICAL ENGINEERING VALUE	TEST METHOD / STANDARD
In-Plane Conductivity (X-Y)	1,000 – 1,500 W/m·K (4× Copper diffusivity)	Laser Flash (ASTM E1461)
Through-Plane Conductivity (Z)	7.0 – 15.0 W/m·K	ASTM E1461
Vacuum Outgassing (No Silicone)	TML < 1.0% | CVCM < 0.10% (Zero oil fogging)	ASTM E595 (125°C, 10^-5 Torr)
Top-Cap Dielectric Isolation	> 5,000 V AC breakdown (0.025 mm Kapton®)	ASTM D149
Composite Total Thickness	0.15 mm – 0.35 mm ultra-low-profile stack	ASTM D374
Graphite Core Density	1.95 – 2.12 g/cm³ (~76% lighter than Copper)	Helium Pycnometry
Operating Temperature Range	-55°C to +150°C (PSA) / +400°C (Core)	MIL-STD-810H Shock
Minimum Bend Radius	2.0 mm dynamic / 1.0 mm static routing	IPC-TM-650 Flex
Flammability Rating	UL 94 V-0	UL 94
Procurement Sovereignty	100% India DVA (< ₹200 Cr MoD GTE Compliant)	Make-in-India Class I/II

InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.

3. ORDERING NOMENCLATURE, PREMIX VARIANTS & COMMERCIAL PRICING SCHEDULE

PART / GRADE CODE	MATERIAL SPECIFICATION & CONSTRUCTION	PROTOTYPE / R&D TIER	VOLUME / OEM TIER
`GFlex-1000 (Std 50×50)`	Edge-Sealed Kapton® + Graphite (50 × 50 mm = 25 cm² ref.)	**₹350.00 Proto | ₹310.00 Pilot**	**₹280.00 OEM Blanket (500+)**
`GFlex-1000 (Large 150×150)`	Large Defense / Space Spreader (150 × 150 mm = 225 cm² ref.)	**₹1,800 Proto | ₹1,650 Pilot**	**₹1,500 OEM Blanket (500+)**
`Space/Defense Kit`	Turnkey Zero-Outgassing Sampler Kit (GFlex + Ultra-NS + ASTM E595)	**₹12,500 / turnkey lab kit**	**Ships in 48 Hours (<₹200Cr GTE)**

4. TARGET INDUSTRIAL APPLICATIONS & DEPLOYMENT SECTORS

- Defense UAV flight computers, phased-array radar Transmit/Receive (T/R) modules, and seeker electronics (BEL, HAL, DRDO, private primes).
- Low-Earth-Orbit (LEO) satellite payloads and optical Earth-observation camera housings requiring zero siloxane outgassing.
- Sealed automotive ADAS LiDAR/camera ECUs and 5G/6G small-cell telecom enclosures with zero vertical clearance for finned heatsinks.
- 800G / 1.6T pluggable optical transceivers in AI data centers routing heat laterally into cold-rail chassis walls.

5. DIGITAL CNC CONVERTING, APPLICATION & STORAGE GUIDELINES

1. Edge-Seal Integrity: Specify perimeter geometry in .DXF CAD so InsulTherm's CNC step-cut creates a 1.0–1.5 mm amber Kapton® dielectric skirt overhang around conductive graphite edges where adjacent SMD components sit within <2 mm. 2. Application: Peel the clear 0.050 mm PET release liner using the kiss-cut tab and apply with 15–30 psi roller pressure onto clean IC/metal surfaces. 3. Cleanroom Option: Available double-bagged in N2-purged ESD pouches for SpaceTech and optical assembly lines.

6. QUALITY ASSURANCE, REGULATORY COMPLIANCE & SOVEREIGN DVA TRACEABILITY

Thermal & Electrical Metrology	Batch-wise ASTM D5470-17 thermal impedance calibration curves (10–50 psi) & ASTM D149 dielectric verification.
Flammability & Fire Safety	UL 94 V-0 rated material formulations; supports ARAI AIS-156 Amendment 3 & UL 9540A pack/rack propagation testing.
Outgassing & Environmental	RoHS 3 (EU 2015/863) & REACH compliant; ASTM E595 low-outgassing trace on Graphite-Flex™ & Ultra-NS™ grades.
Domestic Value Addition (DVA)	100% India Digital CNC Converting (Gujarat); compliant with PM E-DRIVE, SECI BESS DVA & <₹200 Cr MoD GTE Ban.

RAPID 24–48 HOUR CAD-TO-PART PROTOTYPING & OEM PROCUREMENT DESK
Submit .DXF / .STEP drawings, request <₹15,000 Petty-Cash R&D Sampler Kits, or schedule an engineering review:
Primary Official Desk: manav.sb@inslthrm.in | **Active Operations & Order Routing:** manavsb9719@gmail.com
Engineering Helpline: +91 78148 XXXXX | **Manufacturing Wing:** InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India